OFFICIAL BIDDING DOCUMENT

**Telecommunications Technology Solutions Inc. (TelTech)**

Corporate Tender Secretariat · Procurement & Network Engineering Division

**RFP-TELTECH-FWA-2026-001 / MASTER**
**Issuing Body:** Technical Evaluation Board
**Date:** October 14, 2026

# Turnkey 5G FWA Master Tender & RFP Specifications

Nationwide 5,000-Site Turnkey 5G Fixed Wireless Access (Band n50, 60 MHz TDD)

| DOCUMENT NUMBER | ISSUING BODY | TARGET SPECTRUM | BIDDING STATUS |
|---|---|---|---|
| RFP-TELTECH-FWA-2026-001 | Technical Evaluation Board | Band n50 (1.5 GHz, 60 MHz TDD) | Official Request for Proposal |

## 1. EXECUTIVE SUMMARY & INVITATION TO BID

### 1.1 Purpose of the Tender

Telecommunications Technology Solutions Incorporated (TelTech) (hereinafter referred to as the **"Operator"** or **"Company"**) hereby invites qualified, world-class telecommunications equipment manufacturers, network systems integrators, and managed service providers (hereinafter referred to as the **"Bidder"** or **"Vendor"**) to submit comprehensive, binding technical and commercial proposals for the turnkey supply, delivery, installation, integration, optimization, and managed operations of a nationwide **5G Fixed Wireless Access (FWA) Network**. The Operator is actively seeking creative, high-impact proposals—both technical and commercial—from prospective bidders that deliver industry-leading spectral efficiency, innovative vendor financing architectures, accelerated rollout velocity, and optimized Total Cost of Ownership (TCO).

The Operator, a duly franchised telecommunications entity operating pursuant to statutory authority and holding National Telecommunications Commission (NTC) Provisional Authority (PA Case No. 2017-084 for Broadband Data Services and PA Case No. 2017-291 for Nationwide Cellular Mobile Telephone System), has been officially granted a Temporary Permit to Propagate / Demonstrate (Permit No. RB-09-01-2026/DEMO, issued in reference to CPCN No. 2017-084) to conduct test broadcast, propagation, and technology demonstration in Band n50 (1457 – 1517 MHz, 60 MHz channel bandwidth). Furthermore, the Operator holds an active, pending application for the permanent commercial assignment and operational deployment of the said frequency band pursuant to and subject to the provisions of the Open Access in Data Transmission Act (Konektadong Pinoy Law) and applicable NTC regulatory circulars.

### 1.2 Program Scope & Deployment Scale

The nationwide program encompasses a total footprint of **5,000 macro BTS sites** deployed across five (5) consecutive phases:

* **Phase 1 (Day-1 Priority Launch): 1,000 Macro Sites** (Batch 1 — immediate rollout across prioritized unserved and underserved suburban and rural population centers).
* **Phases 2 to 5 (Nationwide Expansion): 1,000 Macro Sites per annum** over Years 2 through 5, reaching full target deployment of 5,000 BTS sites serving over **2.45 to 2.66 Million active broadband subscribers**.
* **Target Addressable Population:** Over **32.3 Million residents** across 5,000 barangays (2024 PSA POPCEN Official Census baseline).
* **Initial Network Sizing Load:** 1,000 active concurrent fixed broadband subscribers per 3-sector BTS (capacity ceiling of 5,965 peak BTS units across expansion).

### 1.3 Tender Structure & Bid Lots

Bidders are invited to submit tenders for one, multiple, or all of the following distinct Lots:
* **LOT 1: Radio Access Network (RAN) Hardware, Antennas & Turnkey Deployment Services**
* **LOT 2: Power Systems & Passive Site Infrastructure (Co-location Interconnect & Hybrid DC Power)**
* **LOT 3: Cloud-Native 5G Core Network (5GC SA) & Datacenter Infrastructure**
* **LOT 4: Converged Digital OSS/BSS, Field Force Automation & Real-Time Charging Engine**
* **LOT 5: RF Planning, Greenfield Site Survey, Civil Works & Cluster Optimization Services**
* **LOT 6: Nationwide Managed Services, 24/7 Tier-2/Tier-3 NOC & Field Network Operations (MS)**
* **LOT 7: Customer Premises Equipment (CPE) 5G Sub-3 GHz High-Gain Terminals**

Bidders submitting integrated "Full Turnkey Consortium" bids spanning Lots 1 through 6 will receive multi-lot synergy evaluation scoring.

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## 2. PROJECT BACKGROUND, OPERATING ENVIRONMENT & TECHNICAL BASELINE

### 2.1 Spectrum Physics: Band n50 Advantage & Operational Channel Plan

* **Authorized Operating Range:** Under the National Telecommunications Commission (NTC) regulatory framework, the authorized operational Band n50 passband is established as **1457.0 MHz to 1517.0 MHz** (a contiguous **60 MHz TDD channel bandwidth** operating at 30 kHz SCS, 162 Resource Blocks).
* **Radio Hardware Tuning Capability:** All active Radio Units (RRU / AAU) and CPE RF front-ends must provide wideband hardware tuning across the full **1432.0 – 1517.0 MHz** Band n50 spectrum (minimum 85 MHz Instantaneous Bandwidth / IBW tuning capability), with active transmission, reception, and sharp digital filtering configured for the authorized 1457.0 – 1517.0 MHz carrier window.
* **Guardband & Satellite Protection Limits:** A 1.0 MHz guardband (1517.0 – 1518.0 MHz) is maintained at the upper boundary. Transceivers must implement sharp filtering ensuring out-of-band emissions into adjacent maritime satellite spectrum (1518.0 – 1525.0 MHz) do not exceed **-30 dBm/MHz** (strictly compliant with ITU-R M.2159-0 / WRC-15 Resolution 223), with Adjacent Channel Leakage Ratio (ACLR) ≥ 45 dB pursuant to 3GPP TS 38.104 Category B.
* **Propagation Efficiency:** 1.5 GHz propagation achieves an effective reliable non-line-of-sight (NLOS) and near-line-of-sight (nLOS) coverage radius of **3.5 km** (delivering up to **38.5 km²** of coverage area per site under suburban/rural clutter).
* **Penetration Resilience:** Foliage, atmospheric rain fade, and typical Philippine residential building materials (concrete hollow block, corrugated galvanized sheet roofing) suffer 8–12 dB lower path loss compared to 3.5 GHz.
* **Frame Structure & Spectrum Allocation:** 60 MHz contiguous TDD frame structure (configurable TDD DL:UL ratios: 8:2 DDDDDDDDSUU, 7:3 DDDDDDDSUU, 4:1 for broadband downlink-heavy traffic, or 3:1 for balanced enterprise transmission).

### 2.2 Site Infrastructure Landscape & Tower Strategy

The operator has completed a nationwide spatial overlay matching the 5,000 candidate barangay centroids against **13,810 nationwide TowerCo registry records (encompassing 13,799 unique Universal Tower IDs and 13,629 distinct geographic tower coordinates)** across Philippine licensed TowerCos:
* **PTCI / MIDC TowerCo:** 5,080 towers
* **FTAP Globe Telecom:** 3,529 towers
* **EdgePoint Infrastructure:** 2,775 towers
* **FTAP Smart Communications:** 1,023 towers
* **LBS Digital Infrastructure (LDIC):** 700 towers
* **FTAP BTS Dedicated Sites:** 639 towers

**Deployment Typology (Co-location vs. Greenfield):**

- **Class T1 (Immediate Co-location, <= 0.5 km):** 54.8% of Phase 1 sites (42.5% nationwide). Sited directly on operational TowerCo structures with existing AC power and civil compound.

- **Class T2 (Close Co-location, 0.5 - 1.0 km):** 26.9% of Phase 1 sites (30.4% nationwide). Sited on operational towers within beamforming reach.
- **Class T3 (Near Co-location, 1.0 - 2.0 km):** 8.0% of Phase 1 sites (17.8% nationwide). Co-location viable subject to RF propagation clearance.
- **Class T4 (Greenfield Monopole Required, > 2.0 km):** 10.3% of Phase 1 sites (9.3% nationwide; ~460 sites total). Requires self-constructed 30m/40m monopole, civil shelter, dedicated utility transformer drop, and backup diesel/solar generation.

### 2.3 Transmission & Backhaul Topology

Transmission connectivity has been audited via the Converge fiber network audit (1,296 confirmed optical nodes and 4,401 fiber segments; subject to verification on the ground):
* **Terrestrial Fiber Optic Cable (FOC):** 98.3% of Phase 1 sites connect via terrestrial fiber drops (Class A: Node <= 0.5 km; Class B: Node <= 1.0 km; Class C: Line-tap splice closure <= 0.5 km; Class D: Lateral/Microwave <= 3.0 km).
* **Satellite LEO Backhaul (Starlink Business):** 1.7% of Commercial Phase 1 sites (and up to 34.4%–43.5% in remote universal access/GIDA sites) utilize 4-terminal Starlink Business arrays (delivering aggregated 400–800 Mbps burst with SD-WAN bonding and traffic shaping).
* **Fiber Baseline Reconciliation Note:** The preliminary raw GIS planning dataset identified 2,405 spatial markers; following ground-truth fiber audit filtering against operational distribution plant, 1,296 confirmed optical nodes and 4,401 fiber segments were verified as immediate interconnect targets (subject to joint site survey verification on the ground).

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## 3. SCOPE OF WORK BY TENDER LOT

### LOT 1: 5G RAN EQUIPMENT & ACCESS HARDWARE

The Bidder shall supply, deliver, install, test, and commission carrier-grade 3GPP Release 16/17 compliant 5G RAN equipment:
1. **Baseband Unit (BBU):**
* High-capacity rack-mountable 19" subrack BBU supporting multi-cell aggregation and up to 12 cells of 60 MHz Band n50.
* Dual redundant DC power supply inputs (-48V DC, operating range -40.5V to -57V DC).
* Backhaul interfaces: Minimum 4x 10GE/25GE optical SFP28 interfaces supporting IEEE 1588v2 Precision Time Protocol (PTP) Telecom Profile (G.8275.1) and SyncE.
* Fronthaul interfaces: Minimum 6x 25GE eCPRI interfaces with integrated Optical Supervisory Channel (OSC).
2. **Remote Radio Units (RRU) / Active Antenna Units (AAU):**
* Configuration: **8T8R MIMO**, 3 sectors per macro site (total of 3 RRUs and 3 antennas per standard macro).
* Operating Frequency: Band n50; full hardware tuning across 1432.0 – 1517.0 MHz (85 MHz IBW capability); active carrier operation and digital filtering configured for the authorized 1457.0 – 1517.0 MHz continuous 60 MHz TDD operating window (60 MHz OBW).
* RF Output Power: Minimum **8 x 40W (320 W total RF power per sector)**.
* Beamforming capability: Digital horizontal and vertical beamforming supporting dynamic multi-user MIMO (MU-MIMO) pairing.
* Protection & Environment: IP65/IP67 ingress rating, natural convection or high-durability fanless cooling, surviving wind speeds of up to 250 km/h (Typhoon Category 5 resilience).
3. **Macro Antennas & RF Ancillaries:**
* High-gain 8-port cross-polarized antennas (or integrated AAU) supporting Band n50 with electrical remote downtilt (eRET via AISG 2.0 / 3GPP).
* Horizontal half-power beamwidth: 65° +/- 5°; Minimum gain >= 17.5 dBi.
* Low PIM performance: <= -153dBc @ 2 x 43 dBm.
* High-grade low-loss optical fiber patchcords, IP67 weatherproof boot assemblies, DC power surge protective devices (SPD Type 1+2), and grounding kits.

### LOT 2: POWER SYSTEMS & PASSIVE INFRASTRUCTURE

The Bidder shall provide unified site power solutions engineered for high availability across both leased TowerCo sites and greenfield sites:

1. **DC Power Rectifier Plant:**

* Modular 19" 3U/4U switchmode DC power system dimensioned for **12.0 kW total installed capacity / 9.0 kW firm N+1 continuous capacity** (4 x 3.0 kW hot-swappable rectifier modules: 3 active + 1 redundant standby in N+1 redundancy).
* Rectifier efficiency: >= 96.0% (80 PLUS Titanium / high-efficiency class).
* Advanced SMPS controller with SNMPv3, HTTPS, Modbus, dry contacts, and remote cloud telemetry.

2. **Battery Energy Storage System (BESS):**

* Lithium Iron Phosphate (LiFePO4 / LFP) battery energy storage system: Sized to deliver at least **12.0 kWh of usable energy** to the specified 3.0 kW continuous DC site load over **4.0 continuous hours** without AC utility power.
* Installed capacity dimensioning must account for maximum 80% usable Depth of Discharge (DoD), conversion/distribution losses, and 80% end-of-life State of Health (SoH) retained capacity (minimum **>= 20.0 kWh nominal installed bank capacity** (minimum 400 Ah to 450 Ah @ 48V, e.g. 4x 48V 105 Ah or 2x 48V 210 Ah, supported by a mandatory loss-based engineering calculation delivering 12.0 kWh net usable energy at 80% DoD, 80% EOL SoH, and DC system losses).
* Bidder shall submit the engineering load calculation, manufacturer discharge curves, and acceptance-test procedure, and demonstrate 4.0-hour autonomy during site acceptance (SAT).
* Cycle life: >= 4,000 cycles at 80% Depth of Discharge (DoD) @ 25°C.
* Built-in intelligent BMS with cell-level balancing, over-charge, over-discharge, short-circuit, and high-temperature thermal runaway cutoffs.

3. **Standby Generator Sets (Greenfield Class T4 Sites):**

* 15 kVA / 12 kW prime-rated, 230V single-phase / 400V 3-phase, 60 Hz water-cooled diesel generator with sound-attenuated weatherproof enclosure (< 68dBA @ 7m).
* Minimum 500-liter to 1,000-liter sub-base day fuel tank providing >= 72hours continuous operation during grid blackout.
* Automatic Transfer Switch (ATS) with motorized 4-pole contactor and mechanical interlock.

4. **TowerCo Co-Location Power Interconnect Kit:**

* Standardized DC tap-off distribution box, sub-metering Class 1.0 revenue-grade energy meter with RS485 telemetry, and dedicated surge suppression.

### LOT 3: CLOUD-NATIVE 5G CORE (5GC SA) & DATACENTER INFRASTRUCTURE

The Bidder shall deploy a carrier-grade, cloud-native 5G Standalone (5G SA) Core Network dimensioned for **1,000,000 initial registered subscribers** scaling up to **3,000,000 subscribers**, deployed in a **Geo-Redundant, Active-Active Dual Datacenter Topology** (NCR Primary DC and Metro Clark / Region III Disaster Recovery DC):

1. **Network Functions (NFs) Scope:**

* Control Plane: AMF, SMF, NRF, NSSF, PCF, UDM, UDR, AUSF, NEF, CHF.
* User Plane: High-performance distributed User Plane Function (UPF) nodes.

2. **User Plane (UPF) Throughput Dimensioning:**

* UPF capacity dimensioned for **80 Gbps usable throughput per 2U dual-socket compute node** (minimum 2 x 100GE line-rate DPDK / SR-IOV interfaces).
* Integrated high-performance **Carrier-Grade NAT (CG-NAT)** supporting deterministic port allocation and 10:1 to 16:1 subscriber port density.
* Integrated Deep Packet Inspection (DPI) for layer-7 traffic classification, fair-use bandwidth throttling, and URL filtering.

3. **Hardware & CaaS Virtualization Infrastructure:**

* Enterprise OpenStack or Bare-Metal Kubernetes (CaaS) orchestration platform.
* Dual-socket Intel Xeon / AMD EPYC server nodes, 512 GB RAM, dual 25G/100G NICs, redundant enterprise NVMe storage arrays.
* Redundant datacenter leaf-spine fabric switches (32 x 100GE spine, 48 x 25GE + 6 x 100GE leaf), redundant perimeter next-gen firewalls, and DDoS mitigation scrubbing appliances.

4. **Licensing Framework:**
* Tiered perpetual subscriber capacity licensing:
* Tier 1 (First 100k subs): Bidder to quote per registered sub.
* Tier 2 (100k to 500k subs): Bidder to quote per registered sub.
* Tier 3 (500k to 1M+ subs): Bidder to quote per registered sub.

### LOT 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION

The Bidder shall deliver an integrated, cloud-native Telecom Business Support System (BSS) and Operations Support System (OSS) optimized for high-volume, low-friction fixed broadband operations:
1. **Converged BSS & Omnichannel Customer Engagement:**
* Omnichannel CRM, Product Catalog, and CPQ supporting 4 primary retail tariffs and SME tiers.
* Native Prepaid-First architecture: Integration directly with 5G Core Charging Function (CHF) via 3GPP Nchf / Diameter Ro interfaces.
* Payment Gateway Integrations: Native, production-ready REST API integrations with Philippine digital payment channels: **GCash, Maya (PayMaya), 7-Eleven CLiQQ OTC, Bayad Center**, bank transfer (InstaPay/PESONet), and municipal distributor airtime portals.
2. **Field Force Management (FMS) & Mobile Technician App:**
* Automated work-order generation and algorithmic dispatch engine handling up to **3,500 daily work orders** (new installs, truck rolls, relocations, CPE reclaims, and trouble calls).
* Android mobile app for field technicians: Real-time GPS dispatch, barcode/QR scanning of CPE MAC and serial numbers, automated outdoor signal sweep tool (RSRP/SINR validation), photograph proof of installation, and customer digital e-signature.
* Turnaround SLA: Compressed installation timeline from 14 calendar days down to **under 48 hours** from order booking.
3. **CPE Warehouse Logistics & Asset Reverse Logistics (WMS):**
* End-to-end serialized asset tracking from central warehouse staging through regional depots, technician truck float, subscriber premises, and repossession/refurbishment cycle.
* Support for minimum 70% CPE recovery and refurbishment upon subscriber churn.
4. **OSS Network Management, Fault & Performance Monitoring:**
* Multi-vendor RAN EMS/NMS integration via 3GPP northbound REST/Kafka/SNMP interfaces.
* Real-time automated fault correlation, root-cause analysis (RCA), and synthetic SLA monitoring.
* Automated NTC regulatory compliance KPI dashboard and QoS reporting engine.

### LOT 5: RF PLANNING, SITE SURVEY & CLUSTER OPTIMIZATION

The Bidder shall execute end-to-end RF planning, site survey, civil design, and cluster radio tuning:
1. **Detailed Greenfield & Co-Location Site Surveys:**
* Technical Site Survey Report (TSSR) for each assigned candidate barangay: Detailed structural load analysis on existing towers, antenna mounting space, azimuthal line-of-sight (LOS) drone photographic sweeps (360° at 20m, 30m, 40m AGL), AC power distribution panel capacity, grounding resistance (<= 5 Ohms), and backhaul cable routing.
2. **High-Precision RF Propagation Modeling:**
* Production-grade RF propagation model tuning in Asset / Atoll / Planet utilizing high-resolution GIS clutter and 3D digital elevation models (DEM <= 5m resolution).
* Cell dimensioning: Sizing for 60 MHz Band n50 TDD, 3 sectors @ 120° nominal separation, antenna height 30m–45m AGL, coverage target RSRP >= -95 dBm across >= 90% of populated barangay settlement areas.
3. **Single Site Verification (SSV) & Cluster Tuning:**
* SSV drive tests and stationary multi-point testing per site verifying 3-sector RF radiation, antenna tilt (RET), PCI validation, azimuth verification, and handover execution.
* Cluster optimization across 10–20 site clusters to eliminate pilot pollution, optimize neighbor lists, minimize inter-cell interference, and balance sector traffic loading.

### LOT 6: NATIONWIDE MANAGED SERVICES & 24/7 NETWORK OPERATIONS (MS)

The Bidder shall provide fully managed, SLA-governed network operations, field maintenance, and spare parts management:

1. **Tier-2 & Tier-3 Network Operations Center (NOC):**

* 24/7/365 active monitoring of RAN, Core, Transmission, Power, and OSS/BSS platforms.
* Incident management, event correlation, dispatching field maintenance teams, and managing customer trouble ticket escalations.

2. **Nationwide First-Line Maintenance (FLM) Field Operations:**

* Regional field maintenance hubs across Greater Luzon, Northern Luzon, Southern Luzon, Visayas, and Mindanao.
* Mean Time to Restore (MTTR) <= 2hours in urban/suburban and <= 4hours in rural clusters for Critical/Severity-1 incidents (with Initial Incident Acknowledgement <= 15 minutes).
* Preventive maintenance: Quarterly inspection of RF antenna brackets, coax/optical jumpers, rectifier modules, battery conductance testing, and generator load testing.

3. **Comprehensive Spare Parts Management (SPMS):**

* Advanced replacement logistics: Central warehouse and regional staging depots maintaining an audited safety float of BBUs, RRUs, rectifiers, lithium batteries, and SFPs.
* Guarantee of next-business-day (NBD) on-site replacement delivery for failed hardware components.

### LOT 7: 5G HIGH-GAIN CUSTOMER PREMISES EQUIPMENT (CPE)

The Bidder shall supply outdoor and high-gain indoor 5G FWA CPE devices:

1. **Indoor 5G High-Gain FWA Router (CPE-01 — 350,000 Units):**

* 3GPP Release 16/17 compliant 5G NR sub-6 GHz processor.
* Frequency Bands: Native Band n50 with wideband tuning across 1432.0 – 1517.0 MHz and optimized operational passband in the authorized 1457.0 – 1517.0 MHz range (60 MHz channel bandwidth) plus fallback sub-6 GHz bands (n1, n3, n7, n28, n41, n77, n78).
* Antenna: High-gain internal omnidirectional multi-element antenna array (minimum >= 8.0 dBi gain across Band n50); 4x4 MIMO DL, 2x2 MIMO UL.
* LAN & Local Wi-Fi: Dual-band concurrent Wi-Fi 6 (802.11ax) AX1800/AX3000, 2.4 GHz (2x2) + 5.0 GHz (2x2) concurrent, OFDMA, MU-MIMO, WPA3-Personal.
* Ethernet: Minimum 1 x 2.5GE + 2 x 1GE RJ-45 LAN ports.
* Remote Management: TR-069 and TR-369 (USP) support with complete TR-181 data model, secure FOTA firmware updates, and Operator PLMN lock (515-XX).

2. **Outdoor 5G High-Gain FWA Terminal Package (CPE-02 — 15,000 Units):**

* High-gain directional integrated antenna array (minimum >= 12.0 dBi gain across Band n50); 4x4 MIMO DL, 2x2 MIMO UL.
* Ethernet: 1 x 2.5GE Power over Ethernet (PoE) input port.
* Enclosure & Durability: IP67 dust and water ingress protection, operating temperature -10°C to +55°C, 200 km/h wind survivability, pole/wall mounting bracket.
* Packaged Companion Bundle: Each outdoor terminal kit includes an indoor Wi-Fi 6 Router/Gateway and 2.5GE-capable 802.3at/bt PoE power injector and indoor router with 2.5GE WAN port.

3. **Staging, Pre-Configuration & SIM Locking (CPE-SVC):**

* Staging, APN configuration, custom operator firmware flashing, PLMN locking, and retail customer kitting covering all 365,000 Phase 1 units.

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## 4. RIGOROUS SERVICE LEVEL AGREEMENTS (SLAs) & PERFORMANCE KPIS

The Bidder must guarantee compliance with the following international telecom best-practice KPIs. Failure to achieve these targets shall trigger contractual liquidated damages (LDs) and service credits.

### 4.1 Radio Access Network (RAN) Performance KPIs

| KPI METRIC | FORMULA / STANDARD | MINIMUM ACCEPTANCE | TARGET EXCELLENCE | MEASUREMENT WINDOW |
|---|---|---|---|---|
| **Radio Network Availability** | [(Total Available Cell Hours - Cell Outage Hours) / Total Available Cell Hours] x 100% | >= 99.70% | >= 99.95% | Monthly / 24x7 |
| **5G Radio Accessibility (CSSR)** | Call/Session Setup Success Rate (RRC + 5G QoS Flow) | >= 96.00% (Min) | >= 97.00% (Target) | Busy Hour (Harmonized w/ Bulletin 02) |
| **5G Session Drop Rate (CDDR)** | Abnormal Release Rate per 100 established sessions | <= 2.00% (Max) | <= 1.50% (Target) | Busy Hour (Harmonized w/ Bulletin 02) |
| **Single Sector User Capacity** | Active concurrent connected CPE sessions per sector | >= 350 CPEs | >= 450 CPEs | Peak Busy Hour |
| **Peak Cell DL Throughput (60 MHz)** | 8T8R 60 MHz TDD peak aggregated sector throughput | >= 450 Mbps | >= 520 Mbps | Lab & Single Cell Test |
| **Cell Edge DL User Throughput** | 5th percentile user speed / 95% coverage threshold (>= 95% of stationary test points) | >= 25 Mbps | >= 50 Mbps | Busy Hour (SSV Governed) |
| **Cell Edge UL User Throughput** | 5th percentile user speed / 95% coverage threshold (>= 95% of stationary test points) | >= 5 Mbps | >= 15 Mbps | Busy Hour (SSV Governed) |
| **End-to-End User Latency (RTT)** | CPE to UPF packet round-trip time | <= 30 ms | <= 18 ms | Continuous Ping |
| **DL Packet Loss Rate (PLR)** | PDCP Layer packet discard rate per 3GPP TS 38.314 | <= 0.20% (Max) | <= 0.10% (Target) | Monthly Average (Harmonized w/ Bulletin 02) |

### 4.2 Power & Site Infrastructure Resilience KPIs

| INFRASTRUCTURE METRIC | TECHNICAL REQUIREMENT | MINIMUM SLA TARGET | FAILURE REMEDY |
|---|---|---|---|
| **Site Power Availability** | BTS Continuous Operating Power (AC Grid + DC Battery + Genset) | >= 99.95% (Leased) >= 99.85% (Greenfield) | Service credit deduction per hour outage |
| **Battery Autonomy Duration** | Time BTS operates on LFP battery during AC utility blackouts | >= 4.0 Hours @ 3.0 kW load | Mandatory module replacement if <3.5 hrs |
| **Generator Auto-Start Reliability** | Genset starts and ATS transfers load within 60s of grid failure | >= 99.0% successful transfers | Liquidated damages on site drop |
| **Grounding Earthing Resistance** | Earth ground electrode resistance to true earth | <= 5.0 Ohms | Mandatory soil conditioning / additional rods |
| **Rectifier System Efficiency** | Operating efficiency from 30% to 90% load range | >= 96.0% | Rejection of non-compliant modules |

### 4.3 5G Core Network & Cloud Infrastructure KPIs

| CORE METRIC | OPERATIONAL DEFINITION | TARGET SLA BENCHMARK |
|---|---|---|
| **5GC Control Plane Availability** | High availability of AMF, SMF, UDM, PCF across geo-cluster | >= 99.999% ("Five Nines") |
| **5GC User Plane Availability** | Availability of UPF packet routing instances | >= 99.999% |
| **PDU Session Establishment Success** | Initial PDU session setup success rate | >= 99.80% |

| CORE METRIC | OPERATIONAL DEFINITION | TARGET SLA BENCHMARK |
|---|---|---|
| **Session Setup Delay** | End-to-end signaling time to establish 5G session | <= 80 ms (95th percentile) |
| **UPF Packet Forwarding Latency** | Processing delay through UPF hardware | <= 1.0 ms |
| **CG-NAT Port Allocation Success** | Real-time port assignment for new outbound sessions | >= 99.99% |
| **Core Failover Recovery Time (DR)** | Automatic cutover from NCR DC to Clark DR DC | <= 30 seconds (Zero session loss across UDM/SMF; user traffic restored <= 30s) |

### 4.4 OSS/BSS, Field Operations & Managed Services SLAs

| OPERATIONAL SERVICE DOMAIN | METRIC DESCRIPTION | MANDATORY SLA BENCHMARK |
|---|---|---|
| **BSS Billing & Rating System Availability** | Online Charging System (CHF/OCS) uptime | >= 99.99% |
| **Payment Top-Up Processing Time** | Time from payment channel confirmation to account credit | <= 3seconds |
| **Field Force Installation Lead Time** | Order placement to physical CPE on-site activation | <= 48 Hours (Standard urban/suburban) |
| **First-Time Right (FTR) Installs** | Installations operating without trouble call within 14 days | >= 95.0% (Harmonized w/ Bulletin 02 & Appendix C) |
| **Customer Churn CPE Recovery Rate** | Successful physical recovery of CPE on churned accounts | >= 70.0% |
| **Severity 1 Incident MTTR (Critical Outage)** | Time to restore full service on macro BTS outage | <= 2 Hours (Urban) / <= 4 Hours (Rural) |
| **Severity 2 Incident MTTR (Major Degraded)** | Time to restore single sector failure or degraded backhaul | <= 4 Hours (Urban) / <= 6 Hours (Rural) |
| **Hardware Replacement Turnaround** | Depot dispatch of spare BBU/RRU/Rectifier to site | Next Business Day (NBD) nationwide (<= 24h for Metro Manila/regional hubs) |

## 5. TENDER SUBMISSION STRUCTURE & COMPLIANCE REQUIREMENTS

Bidders must submit their proposals divided strictly into three (3) separate volumes:

### Volume I: Executive & Commercial Proposal (Financial Envelope)

1. **Form A — Bid Submission Letter & Consortium Structure:** Executed by authorized corporate officers.
2. **Form B — Bill of Quantities (BoQ) & Unit Price Book:** Exhaustive pricing matrix in USD and PHP for all hardware, software licenses, implementation services, and annual maintenance contracts (AMC), completed using the official blank Excel BoQ template.
3. **Form C — Total Cost of Ownership (TCO) Breakdown (10-Year Horizon):** Itemized Day-1 CAPEX, expansion CAPEX, and 10-year OPEX schedules.
4. **Form D — Vendor Financing Proposal (Mandatory Requirement):**
5. The Bidder **must** provide a detailed vendor financing facility covering at least **80% of Capex** (BTS hardware, core, and turnkey rollout).
6. Tenor: Flexible facility tenor up to 20 Years (minimum 4 Years, 5+ Years Preferred).
7. Interest Rate: Fixed commercial rate (preferred <= 6.50% p.a.) or competitive benchmark to SOFR + spread.
8. Deferred Payment / Grace Period: Deferred principal repayment / grace period up to 10 Years (minimum 12 Months, 18+ Months Preferred).

### Volume II: Technical Architecture & Compliance Matrix (Technical Envelope)

1. **Section A — RAN Technical Design:** 3GPP R16/17 compliance, Band n50 8T8R AAU/RRU datasheets, BBU subrack capacity, antenna radiation patterns, and link budget calculations.
2. **Section B — Power & Site Civil Engineering:** Rectifier plant, LFP battery cycle curves, genset specifications, and TowerCo co-location interface diagrams.
3. **Section C — 5G Core Architecture:** Cloud-native microservices architecture, dual datacenter active-active geo-redundancy topology, UPF DPDK hardware dimensioning, and CG-NAT sizing.
4. **Section D — OSS/BSS & IT Stack:** Modular architecture, digital self-care portal, payment channel API integrations, field-force management mobile application, and WMS inventory tracking.
5. **Section E — RF Planning & Cluster Optimization Methodology:** Link budget models, clutter definitions, drive-test methodology, and SSV acceptance criteria.
6. **Section F — Managed Services & Operations Plan:** NOC structure, FLM staffing model, SLA penalty framework, and spare parts dimensioning model.
7. **Compliance Matrix:** Line-by-line statement of compliance (**Compliant / Minor Deviation / Non-Compliant**) against every technical requirement in Appendix A.

### Volume III: Bidder Qualifications, References & Financial Soundness

1. **Corporate Financial Statements:** Audited financial statements for the past three (3) fiscal years demonstrating solvency, liquidity, and operational turnover.
2. **Telecom Track Record:** Proof of at least three (3) commercial 5G deployments (RAN or Core) of similar or larger scale within the past 36 months.
3. **Philippine In-Country Support Structure:** Documented local Philippine engineering entity, SEC registration, local repair center, and existing field logistics capabilities.

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## 6. BID EVALUATION CRITERIA & AWARD METHODOLOGY

Tenders will be evaluated using a **Quality and Cost-Based Selection (QCBS)** scoring framework:

```
Total Evaluation Score = (Technical Score × 60%) + (Commercial & Financing Score × 40%)
```

### 6.1 Technical Evaluation Breakdown (100 Points / 60% Total Weight)

- **RAN Architecture & Radio RF Performance (25 Points):** Band n50 output power, 8T8R beamforming algorithms, power consumption efficiency, and environmental durability.
- **5G Core Architecture & Geo-Redundancy (20 Points):** Cloud-native architecture, UPF line-rate throughput, CG-NAT scale, and seamless hitless failover.
- **OSS/BSS Capabilities & Local Integrations (15 Points):** Real-time charging, native Philippine digital payment interfaces, automated field dispatch tool, and WMS inventory reverse logistics.
- **Power Engineering & Site Solution (10 Points):** LFP battery cycle life, rectifier efficiency, TowerCo co-location integration simplicity, and genset reliability.
- **RF Planning, Optimization & Rollout Methodology (15 Points):** Link budget precision, cluster tuning rigor, and proven ability to mobilize 1,000 sites in 12 months.
- **Bidder Experience, References & Local Entity (15 Points):** Tier-1 telecom reference base, local Philippine support team, and spare parts warehouse infrastructure.

### 6.2 Commercial & Financing Evaluation Breakdown (100 Points / 40% Total Weight)

- **Capex Competitiveness (40 Points):** Evaluated across lowest compliant Capex bids. Lowest qualifying responsive Capex receives full points.

- **10-Year Opex & TCO Efficiency (25 Points):** Low ongoing power consumption, competitive software AMC rates, and managed services unit pricing.
- **Vendor Financing Structure (35 Points):** Grace period length, interest rate spread, financing quantum (>= 80%), and ease of financial covenants.

## 7. TENDER TIMELINE & KEY MILESTONES

| MILESTONE EVENT | SCHEDULED DATE | RESPONSIBILITY |
|---|---|---|
| **Official RFP Tender Issuance** | October 14, 2026 | Operator Tender Committee |
| **Bidder Registration & NDA Signing** | October 21, 2026 | Prospective Bidders |
| **Pre-Bid Clarification Meeting (Hybrid)** | October 28, 2026 | Operator & Bidders |
| **Deadline for Written Inquiries / RFIs** | November 4, 2026 | Prospective Bidders |
| **Operator Formal Addendum & Clarifications** | November 11, 2026 | Operator Tender Committee |
| **Tender Submission Deadline (Sealed Bids)** | December 2, 2026 (17:00 PHT) | Bidders |
| **Opening of Technical Envelopes** | December 4, 2026 | Operator Technical Board |
| **Technical Clarifications & Vendor Presentations** | December 7 – 14, 2026 | Shortlisted Bidders |
| **Opening of Commercial & Financing Envelopes** | December 16, 2026 | Operator Commercial Board |
| **Final Contract Negotiation & Award** | December 22, 2026 | Operator & Winning Bidder(s) |
| **Phase 1 Contract Execution & Kickoff** | January 15, 2027 | Operator & Appointed Contractor |
| **Phase 1 Target Commercial Launch (1,000 Sites)** | Target Q4 2027 (Contractual completion within 365 calendar days from NTP / 15 January 2028) | Joint Deployment Team |

## 8. INSTRUCTIONS TO BIDDERS & SUBMISSION DETAILS

### 8.1 Clarifications and Communications

All formal communications, requests for interpretation, and tender queries must be submitted in writing via official email to:
* **Tender Secretariat:** `oliver.tungol@comclark.com.ph`
* **Attention:** Technical Evaluation Board
* **Subject Line:** `[RFP-TELTECH-FWA-2026-001] Technical Clarification Request - <Bidder Name>`

### 8.2 Submission Format & Channels

- **Definitive Electronic Submission (Primary Channel):** Bidders must submit all technical compliance matrices, commercial BoQ schedules, financing term sheets, and required credential attachments 100% digitally through the dedicated **TelTech 5G FWA E-Tender Portal** (https://quizoli.github.io/fwa-5g-presentation/rfp_login.html) on or before **December 2, 2026, at 17:00 PHT (09:00 UTC)**. Upon successful submission, the portal records an authoritative cryptographic server receipt with SHA-256 seal.

- **Contingency Encrypted Backup:** In the event of an extraordinary telecommunication failure or portal network interruption, bidders may transmit an AES-256 encrypted archive to the Tender Secretariat (`oliver.tungol@comclark.com.ph`) prior to the submission deadline. Encryption keys for commercial files must be withheld until the formal commercial opening date.
- **Elimination of Paper Submissions:** To streamline evaluation, eliminate administrative friction, and maintain verified digital provenance, **no physical paper hardcopy or USB flash drive delivery is required** for initial proposal submission. Hardcopy execution copies will be required solely from the awarded contractor(s) upon Notice of Award.

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*(End of Request for Proposal Specification — Tender Appendices Follow)*

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